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3D Optimal Surveillance Trajectory Planning for Multiple UAVs by Using Particle Swarm Optimization With Surveillance Area Priority

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ABSTRACT The use of the unmanned aerial vehicle (UAV) has been regarded as a promising technique in both military and civilian applications. However, due to the lack of relevant laws and regulations, the misuse of illegal drones poses a serious threat to social security. In this paper, we develop a trajectory planner based on particle swarm optimization and a proposed surveillance area importance updating mechanism aimed at deriving three-dimensional (3D) optimal surveillance trajectories for multiple monitoring drones. We also propose a multi-objective fitness function in accordance with energy consumption, flight risk, and surveillance area priority in order to evaluate the trajectories generated by the proposed trajectory planner. Simulation results show that the trajectories generated by the proposed trajectory planner can preferentially visit important areas while obtaining a high fitness value in various practical situations.

INDEX TERMS Particle swarm optimization, 3D path planning, surveillance area priority, multiple unmanned aerial vehicles.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs), also known as drones, have become increasingly important in both military and civilian applications over the past few decades in areas such as remote sensing [1], sensor data collection [2], [3], UAV-enabled communications [4], [5], relays for ad hoc networks [6], disaster monitoring [7], flood area surveillance [8], and wildfire tracking [9]. However, due to a lack of proper laws and regulations, the misuse of drones poses a serious threat to public safety. Since monitoring drones (MDrs) are usually powered by a battery, feasible surveillance trajectories for MDrs should be carefully designed considering their physical restrictions. As a fundamental element of an UAV autonomous control module, the UAV trajectory planning problem has been studied for decades. It can be formulated as an optimization problem that finds the feasible path from the source to the destination. And the optimal trajectory is usually associated with the path that maximizes (or minimizes) a certain optimization index (e.g., energy consumption, path length, etc.) of a certain mission.

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A. RELATED WORK

To derive the optimal trajectory, many researchers have proposed numerous trajectory planners. Turker *et al.* proposed a path planner based on a simulated annealing algorithm to obtain a nearly optimal path in a two-dimensional (2D) radar-constrained environment [10]. In [11], Yoo *et al.* utilized the A* algorithm to derive the optimal UAV trajectory to collect sensing data in a wireless sensor network. However, those proposals (designed for a 2D environment) failed to be applied in 3D operational space, since more constraints need to be modeled to acquire the optimal trajectories. For 3D trajectory planning, algorithms like D* [12], rapidly exploring random tree (RRT) [13], bio-inspired algorithms [14], [15], or an evolutionary algorithm (EA) [16]–[19] are used.

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Since finding the optimal solution to the trajectory problem is non-deterministic polynomial time–complete [20], EAs are the best optimizers due to their advantages when dealing with highly complicated 3D trajectory planning problems. In [18], a state-of-the-art variant of a differential evolutionary (DE) algorithm was employed to solve the 3D trajectory planning problem in a snagged environment. For multiple UAV path planning, Besada-Portas *et al.* [17] proposed a trajectory planner based on a multiple coordinated agents coevolution EA (MCACEA), in which the optimization criteria include 11 optimization indexes and constraints. Li *et al.* [21] proposed a variable neighborhood descend (VND) enhanced genetic particle swarm optimization (PSO) trajectory planner for multiple UAVs in an agricultural application scenario. However, the operational time and the path length are the only optimization indexes in their scheme. Differences in system methodologies in the literature results in difficult to compare the proposed strategy with the other researches. In the literature, there are several techniques based on acoustic data for feature extraction, such as harmonic line association [22], [23], the wavelet transform [24], and the mel-frequency cepstral coefficient (MFCC) [25] method. The second step is classification, and for this, many mathematical models can be used, such as the support vector machine (SVM) [26], the Gaussian mixture model [27], and the hidden Markov model (HMM) [28]. Kaleem and Rehmani presented schemes for drone localization and tracking [29]. Therefore, it is very difficult to compare the proposed acoustic-based scheme for positioning and tracking of illegal drones strategy with the other researches. Hence, different philosophy and targets of system designs in [10]–[29] result in totally different system parameters. For comparative performance analysis, we are unable to compare our approach with the existing literature because still there is no work is available in the literature like ours scenario. Hence, it is not suitable to compare the performances due to different system methodologies. Unlike the resource-allocation and interference-mitigation schemes [30]–[46], this paper addresses three dimension optimal surveillance trajectory planning for multiple UAVs by using PSO with surveillance area priority.

B. MAIN CONRTRIBUTIONS

The objective of this paper is to derive the optimal surveillance trajectories for multiple monitoring drones to surveil a certain operational area, and to detect the existence of illegal drones (IDrs). To solve this problem, we propose a trajectory planner based on PSO and surveillance area priority. Moreover, we extend our trajectory planner to a 3D environment. Using the proposed trajectory planner, the optimal trajectories can be obtained from all possible trajectories in accordance with the proposed fitness function. In our proposed multi-objective fitness function, not only the energy consumption (EC) but the UAV maneuverability, flight risk, and surveillance area priority are also jointly considered cost-determinant. Taking into consideration all these aspects make our approach more practical in UAV trajectory planning.

The rest of the paper is organized as follows. Section II presents the problem description n, along with explanations of terrain and trajectory representation. A multi-objective fitness function for trajectory optimization is introduced in Section III. Then, the proposed surveillance area priority updating mechanism is presented in Section IV, and the trajectory planner is described in Section V. In Section VI, the simulation results and performance analysis of the proposed trajectory planner are illustrated in detail. Finally, we provide concluding remarks in Section VII.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. PROBLEM DESCRIPTION

As shown in Figure 1, MDrs with similar specifications are employed to surveil the whole operational are local maxima to detect the existence of IDrs in 3D operational space. During the surveillance, we stipulate that the MDrs cannot enter any area where prohibited by regulations. To avoid being destroyed as a hostile drone, the MDr also cannot approach the ground-based detection system (GBDS) areas equipped for ground-based drone detection. Furthermore, we assume the MDrs communicate with each other by UAV-to-ground link or in an ad-hoc manner. Therefore, MDrs can mutually share information during execution of the flight task.

In our implementation, we discretize the whole operational area into several small unit areas called cells, as shown in Figure 2, in which the area in red represents the restricted area. We assume that an MDr can cover four cells from a certain position (i.e., a waypoint) depending on the coverage slope of the camera imaging sensor on board (i.e., the areas marked in blue).

B. TERRAIN REPRESENTATION

To mimic a real-life terrain, we adopt a variant of the Foxhole Shekel function (Figure 3) in our paper to represent the landscape, which is formulated as expressed in [47] (1):

$$F(X) = \sum_{i=1}^{10} \frac{0.1}{\sum_{j=1}^{10} (x_j - \eta_{ij})^2 + \gamma_i} \tag{1}$$

where parameters η and γ are utilized to vary the terrain shape. Due to the lack of widely accepted benchmarks in the field of trajectory planning for UAVs, we adopted this terrain because the local maxima of the landscape can be considered mountains.

C. TRAJECTORY REPRESENTATION

In our implementation, the trajectories generated by the optimization algorithm are a sequence of three-dimensional waypoints. Therefore, a feasible path is encoded as a vector where the element $\boldsymbol{w}_i = (x_i, y_i, z_i)$ represents the i-th waypoint, as shown in (2):

$$Trajectory = (\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_{N_w}) \tag{2}$$

where N_w is the number of waypoints in a feasible trajectory.

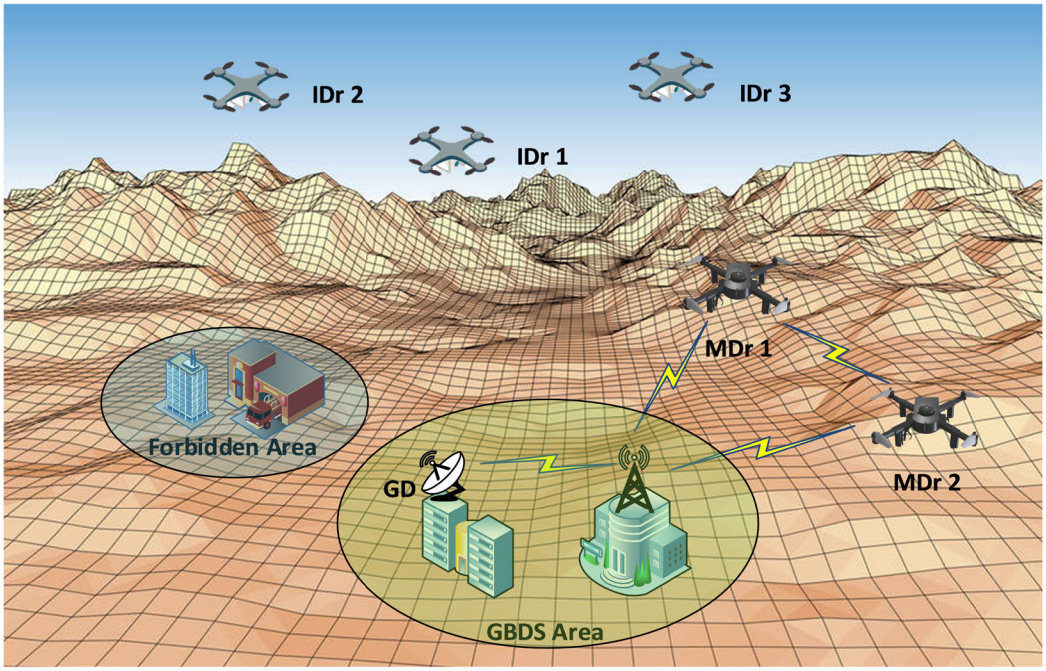


FIGURE 1. Operational scenario.

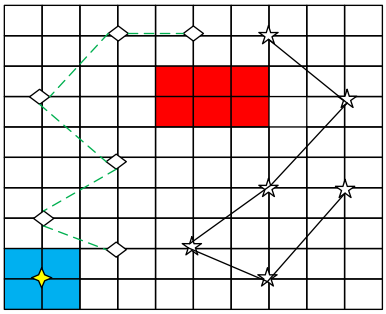


FIGURE 2. Example operational area and waypoints of MDrs.

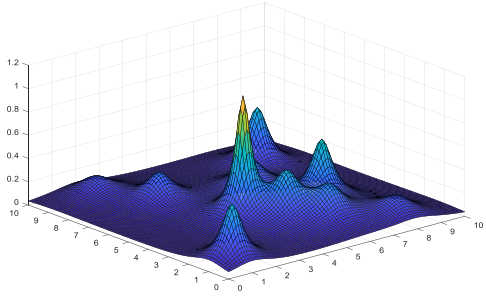


FIGURE 3. Representation of the terrain under surveillance.

III. FITNESS FUNCTION FOR UAV TRAJECTORY OPTIMIZATION

In this section, to evaluate the trajectories generated by the proposed multi-UAV path-planning algorithm, we propose a multi-objective fitness function that consists of eight optimization indexes. To emphasize the importance of different optimization indexes during the optimization process, we divide them into two groups and assign different priority levels: (I) the constraints (terrain, forbidden areas, turning angle, flying slope, and multi-UAV collision avoidance) that the UAV must satisfy due to its physical limitations, and (II) the optimization objectives (energy consumption, flying risk, and surveillance area importance) that must be maximized according to certain mission criteria.

Table 1 shows these classifications and the equations to calculate them. For all possible UAV trajectories, the one with the higher fitness value is always preferred. Therefore, we formulate the fitness function as:

$$F_{fitness} = F_{objective} + TC + FAC + TAC + FSC \tag{3}$$

where $F_{objective}$ is the objective function for which we need to maximize the value in order to derive an optimal trajectory. The rest of the parts correspond to the constraints that should be satisfied before planning a trajectory. More details will be explained in the following sections.

A. OBJECTIVE FUCNTION DESGING

One of the optimization criteria is the objective function, which is used to improve the quality of trajectory planning. We define the objective function as a weighted component of energy consumption, flight risk, and surveillance area importance. So, the objective function is formulated as expressed in (4):

$$F_{objective} = -w_1 F_{EC} - w_2 F_{FR} + w_3 F_{SAI} \tag{4}$$

where F_{EC}, F_{FR}, and F_{SAI} are defined in the range $[0, 1]$, and $w_i (i = 1, 2, 3)$ is the weight of each component, which

TABLE 1. Classifications for optimization indexes.

I. Constraints					
Name	Terrain	Forbidden Area	Turning Angle	Flying Slope	Collision Avoidance
Abbreviation	TC	FAC	TAC	FSC	CAC
Equation	(17)	(18)	(19-20)	(21-22)	(23)
Priority level	1st	1st	1st	1st	1st
Value/Range	0	0	0	0	0
II. Optimization Objectives					
Name	Energy Consumption		Flight Risk	Surveillance Area Importance	
Abbreviation	EC		FR	SAI	
Equation	(5-9)		(10-13)	(14-16)	
Priority level	2nd		2nd	2nd	
Value/Range	[0,1]		[0,1]	[0,1]	

reflects the important differences while evaluating a candidate path. Intuitively, a path with less energy and flight risk, but a higher surveillance area importance value, is always preferable. So, we stipulate energy consumption and flight risk as negative values, whereas surveillance area importance is positive.

1) ENERGY CONSUMPTION

Small drones are usually powered by a battery, which means they must finish the surveillance task before consuming all their energy. Thus, a feasible path with lower fuel consumption is always preferred. We assume the UAV velocity is constant during the operational time. The EC can be formulated as follows:

$$F_{EC} = \frac{\sum_{i=1}^{N_w-1} EC_i}{\max EC} \tag{5}$$

$$EC_i = P_u \times t_{i,i+1} \tag{6}$$

$$t_{i,i+1} = \frac{d_{i,i+1}}{v} \tag{7}$$

$$d_{i,i+1} = \sqrt{(x_{i+1}-x_i)^2 + (y_{i+1}-y_i)^2 + (z_{i+1}-z_i)^2} \tag{8}$$

where EC_i is the fuel consumption from the i'th waypoint to the $(i+1)$'th waypoint. P_u is the energy consumption at velocity v for the time unit; $t_{i,i+1}$ is the flight time from the i 'th waypoint to the waypoint $(i+1)$; $d_{i,i+1}$ is the 3D flight cartesian distance between the i'th waypoint and waypoint $(i+1)$; and $\max EC$ is the normalized constant value, formulated as:

$$\max EC = (N_w - 1) \times P_u \times \frac{d_{\max}}{v}, \quad d_{\max} = \sqrt{X^2 + Y^2 + Z^2} \tag{9}$$

where, the X, Y, and Z representing the x-axis, y-axis, and z-axis, respectively.

2) FLYING RISK

The physical characteristics (e.g., small and lightweight) of the MDr make it susceptible to weather conditions (e.g., rain, snow) during the surveillance task. In addition, flying altitude can be another big risk, since a higher altitude means stronger winds in which the MDrs may be accidentally destroyed. Based on the above scenario, we define the following two kinds of flight risk (FR).

a: ENVIRONMENTAL RISK

Due to the strong random characteristics of environmental risk, it is difficult to build a precise mathematical model. For simplicity, we randomly generated an environmental risk value for each waypoint. And the environmental risk, $r^e_{i,i+1}$, between the i'th waypoint and waypoint $(i+1)$ is defined as the sum of their environmental values.

b: FLYING ALTITUDE RISK

The flying altitude risk is proportional to the absolute flying altitude difference between each of two waypoints. Therefore, we formulate flying altitude risk $r^a_{i,i+1}$ as expressed in (10):

$$r^a_{i,i+1} = \chi \times (z_{i+1} - z_i) \tag{10}$$

where χ is a constant control parameter.

Flying risk is a location-dependent parameter, and it increases or decreases only depending on the weather conditions and UAV flying altitude during the flight. The total flying risk can be computed as seen in equations (11) to (13):

$$F_{FR} = \frac{\sum_{i=1}^{N_w-1} FR_i}{\max FR} \tag{11}$$

$$FR_i = w_{ER} r^e_{i,i+1} + w_{AR} r^a_{i,i+1}, \; w_{ER} + w_{AR} = 1 \tag{12}$$

where FR_i is the flight risk from the i'th waypoint to the waypoint $(i+1)$, and w_{ER} and w_{AR} are the weights of the environmental risk and the flight altitude risk, respectively; $\max FR$ is the normalized constant value, which is written as:

$$\max FR = (N_w - 1) \times \left[w_{AR} \times Z + w_{ER} \times (2 \times \max r^e) \right] \tag{13}$$

where $\max r^e$ represents the maximum environmental risk and Z represents the flying altitude.

3) SURVEILLANCE AREA IMPORTANCE

When an MDr is assigned to execute a certain surveillance task, we want it to first surveil important areas, i.e., to obtain higher surveillance area importance (SAI) values. Thus, we introduce SAI values to characterize different surveillance area priorities among the cells. In our implementation, we assign a random SAI value to each cell, which applies to the whole grid on the map. The random assignment of SAI values means that some cells have higher SAI values while the others have lower SAI values. So, the normalized SAI value of a feasible trajectory can be calculated by equations (14) to (16):

$$F_{SAI} = \frac{\sum_{i=1}^{N_w} SAI_i(t)}{\max SAI} \tag{14}$$

$$SAI_i = \sum_{cell\ x\ \in N(i)} v_{cell\ x}(t) \tag{15}$$

$$\max SAI = N_w \times N_n \times V_{\max} \tag{16}$$

where $SAI_i(t)$ and $v_{cell\ x}(t)$ are the SAI values of the i-th waypoint and of cell x for flight time t, respectively. $N(i)$ is the cell set supervised from the i-th waypoint. N_n is the number from $N(i)$, and $V_{\max}$ is the maximum SAI value.

B. CONSTRAINT FUCTION DESIGN

The constraint functions are used to evaluate the feasibility of a path. When they are satisfied, each constraint is equal to 0, but a negative penalty value, Q, is chosen when they are not. By choosing a value for Q that is less than -1, we can guarantee that the fitness values of feasible paths are always greater than any unfeasible ones. Therefore, we can always obtain a feasible trajectory if all constraint functions are satisfied during the optimization process.

1) TERRAIN CONSTRAINT

An MDr cannot literally go through the terrain (e.g., collide with mountains). Thus, the flying altitude of an MDr must be higher than the terrain's altitude. We use terrain function $Altd(x, y)$, explained in Section III, to determine the altitude of any position (x, y). Then the terrain constraint can be described as:

$$TC = 0,\quad TC = \sum_{i=1}^{Nw} TC_i$$

$$where\ TC_i = \begin{cases} Q, & if\ z_j < Altd(x_i, y_i) \\ 0, & otherwise \end{cases} \tag{17}$$

2) FORBIDDEN AREA CONSTRAINT

For some specific areas (e.g., sensitive government regions), the MDr cannot enter due to regulations. A legal path should be carefully designed to avoid those restricted areas. For the sake of simplicity, we assume that those forbidden areas are rectangles. The forbidden area constraint (FAC) can be formulated as:

$$FAC = 0,\quad FAC = \sum_{i=1}^{Nw} FAC_i$$

$$where\ FAC_i = \begin{cases} Q, & if\ waypoint\ i\ in\ Range(x_j, y_j) \\ 0, & otherwise \end{cases}$$

$$where\ Range(x_j, y_j) = \left\{l_x \le x_j \le u_x\right\} \cap \left\{l_y \le y_j \le u_y\right\} \tag{18}$$

where l_x and l_y are the lower bounds of the x and y coordinates, respectively, of the j-th forbidden area, and u_x and u_y are the upper bounds of the x and y coordinates, respectively, of the j-th forbidden area.

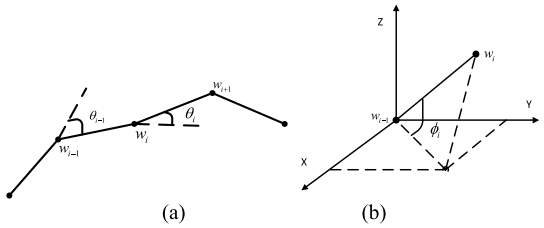


FIGURE 4. UAV maneuverability parameters: (a) turning angle, and (b) flying slope.

3) TURNING ANGLE CONSTRAINT

The turning angle is defined as the horizontal angle between the previous and current directions (as seen in Figure 4). A practical path should be adequately smooth for the UAV to maneuver through easily. Therefore, the turning angle of the UAV is required to be less than the maximum tolerant turning angle. This constraint can be formulated with (19):

$$TAC = 0,\ TAC = \sum_{i=2}^{Nw-1} TAC_i,$$

$$where\ TAC_i = \begin{cases} Q, & if\ \theta_i > \theta_{\max} \\ 0, & otherwise \end{cases} \tag{19}$$

where θ_i is the turning angle at the i-th waypoint (x_i, y_i, z_i), and $\theta_{\max}$ is the maximum tolerable turning angle. In [16], Zheng at al. suggested the formulation of θ_i is (20), as shown at the bottom of this page, where $\|x\|_2$ is the norm of vector x.

4) FLYING SLOPE CONSTRAINT

Analogous to the turning angle in the horizontal direction, we introduce *flying slope* to indicate UAV maneuverability

$$\theta_i = \arccos\left(\frac{(x_i - x_{i-1},\ y_i - y_{i-1}) \cdot (x_{i+1} - x_i,\ y_{i+1} - y_i)^T}{\|(x_i - x_{i-1},\ y_i - y_{i-1})\|_2 \cdot \|(x_{i+1} - x_i,\ y_{i+1} - y_i)\|_2}\right) \tag{20}$$

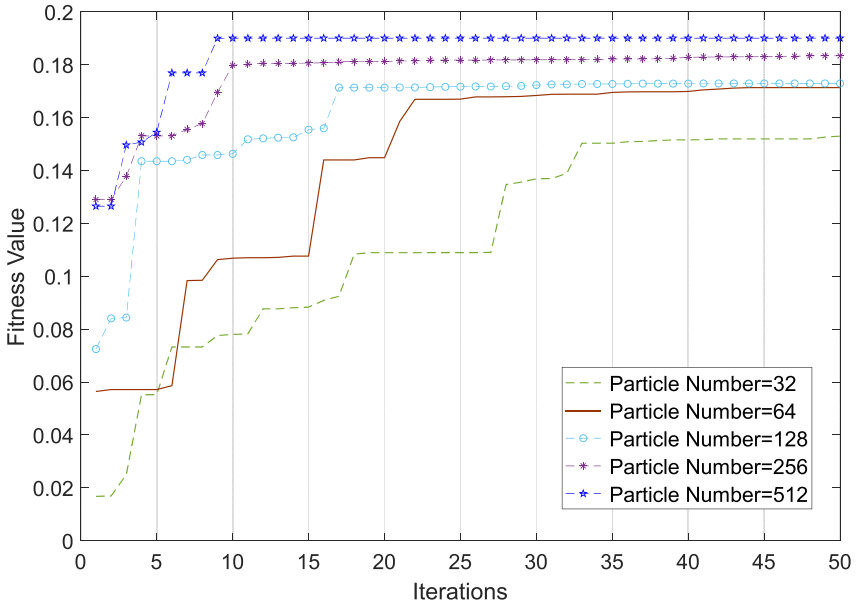


FIGURE 5. Optimal fitness values for the different numbers of particles and iterations.

in the vertical direction, i.e., gliding and climbing angle. The flying slope is defined as the slope between the horizontal direction of the current waypoint to the next one (as shown in Figure 4). The flying slope must be within the scope of the maximum gliding and climbing angles. The flying slope constraint (FSC) can be formulated with (21):

$$FSC = 0, \quad FSC = \sum_{i=2}^{Nw} FSC_i$$
$$where\ FSC_i = \begin{cases} Q, & if\ r_i \notin [\tan(\alpha_{\max}),\ \tan(\beta_{\max})] \\ 0, & otherwise \end{cases} \tag{21}$$

where $\alpha_{\max}$ and $\beta_{\max}$ are the maximum tolerable gliding and climbing angles, respectively, and r_i is the flying slope at the i'th waypoint (x_i, y_i, z_i). Zheng *et al.* [16] suggested a formulation of it as seen in (22):

$$r_i = \frac{z_i - z_{i-1}}{\|(x_i - x_{i-1}, y_i - y_{i-1})\|_2} \tag{22}$$

5) MULTIPLE UAV COLLISION AVOIDANCE CONSTRAINT

Our focus is on trajectory planning for multiple UAVs. When multiple UAVs are used for a complex surveillance mission, the paths should be carefully designed for the drones in order to avoid collisions among them, which can be vital for task implementation. For two separate trajectories, UAVs should maintain a minimum safe distance between them to avoid collisions. This constraint can be described as follows:

$$CAC = 0, \quad CAC = \sum_{i=1}^{N_w^p} \sum_{j=1}^{N_w^q} CAC_i$$
$$where\ CAC_i = \begin{cases} Q, & if\ d_{ij}^{mn} < d_{\min} \\ 0, & otherwise \end{cases}$$
$$d_{ij}^{mn} = \sqrt{(x_j^m - x_i^n)^2 + (y_j^m - y_i^n)^2 + (z_j^m - z_i^n)^2} \tag{23}$$

where $d_{\min}$ is the minimum safe distance to avoid collisions, d_{ij}^{uv} is the cartesian distance between the i-th waypoint of the u-th UAV trajectory and the j-th waypoint of the p-th UAV trajectory.

IV. PROPOSED MCHP-RA SCHEME SURVEILLANCE AREA PRIORITY UPDATING MECHANISM BASED ON EVENT DETECTION

In this section, we propose a surveillance area priority updating mechanism based on the detection of an event. When an MDr detects the existence of IDrs in a certain area (i.e., an event is detected in the area), on intuition, it will be more attentive to that area during the next flight, or it shares the information with other MDrs. Additionally, an MDr should also pay more attention to the areas where SAI values are rapidly changing compared to their historical average SAI values. Moreover, it is necessary to increase the SAI values for those areas that have not been surveilled for more than a certain number of flights so that MDrs can cover them during the next flight. To increase surveillance efficiency, we reduce the SAI values for those areas that were previously surveilled to avoid repeated surveillance of the same areas. Based on the above requirements, we define four cases for updating SAI values.

- When IDrs are detected in a certain cell, surveil its neighboring cells to prevent IDrs from intruding into those areas. Therefore, increase the SAI values of that cell as well as its neighboring cells.
- If the SAI value of one cell changes rapidly compared to its previous average historical SAI values, then the SAI value of that cell is adjusted.
- When a cell has not been surveilled for more than a specific flight times, we should increase its SAI value.
- If a cell has been surveilled, we should reduce its SAI value.

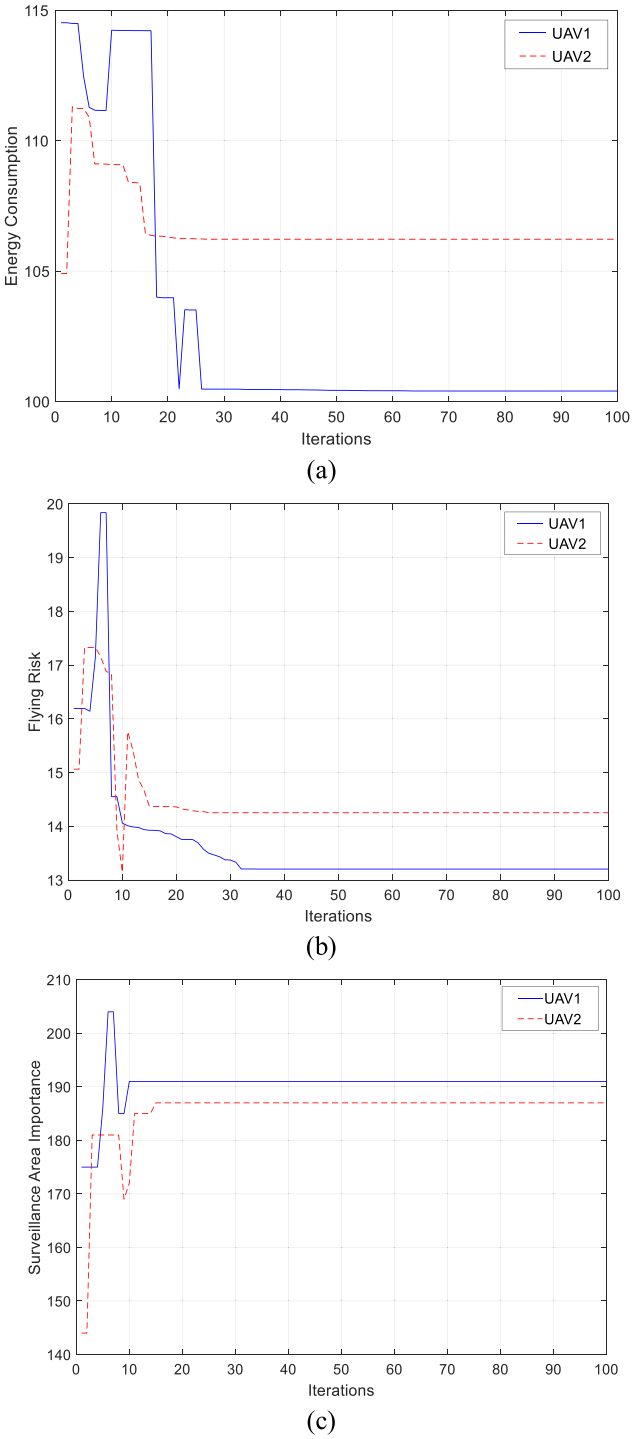


FIGURE 6. PSO trajectory optimization performance in terms of (a) energy consumption, (b) flight risk, and (c) surveillance area importance (SAI).

Based on the above descriptions, we formulate those cases as follows.

For the first case, when $|SAI_i(t) - SAI_i(t-1)| > th_{event}$, then an event has happened in those cells that have been supervised from the i'th waypoint, and the SAI values of those cells can be updated by using (24):

$$v_{cell\ x}(t) = \max[v_{\max},\ \gamma_1 \times v_{cell\ x}(t-1)],\quad \gamma_1 > 1 \quad (24)$$

where $SAI_i(t)$ is the SAI value of the i'th waypoint during the t'th flight time, $v_{cell\ x}(t)$ represents the SAI value of cell x during the t'th flight time, and th_{event} and $v_{\max}$ are the event detection threshold and the maximum SAI value, respectively.

For the second case, when the absolute value between the current SAI value of cell x and its previous historical average SAI value is more than the predetermined updating threshold, th_{update}, then its SAI value is updated by the value returned by $\max[v_{\max}, f_{cell\ x}(t)]$, so we formulate the second case with (25) and (26):

$$v_{cell\ x}(t) = \begin{cases} v_{cell\ x}^{ini}, & if\ \left|\overline{V_{cell\ x}(t-2)} - v_{cell\ x}(t-1)\right| < th_{update} \\ \max[v_{\max}, f_{cell\ x}(t)], & otherwise \end{cases} \quad (25)$$

$$f_{cell\ x}(t) = \log(\lambda \times \left|\overline{V_{cell\ x}(t-2)} - v_{cell\ x}(t-1)\right| + 1) \times v_{cell\ x}(t-1) \quad (26)$$

where $\overline{V_{cell\ x}(t-2)}$ represents the average SAI value from the first flight time to $(t-2)$ th flight time, $v_{cell\ x}^{ini}$ is the initial SAI value of cell x, and λ is the constant control parameter.

For the third case, when a cell has not been surveilled for more than a certain flight times threshold th_{flight}, then we increase its SAI value using (27):

$$v_{cell\ x}(t) = \max[v_{\max}, \gamma_2 \times v_{cell\ x}(t-1)],\quad \gamma_2 > 1 \quad (27)$$

For the fourth case, when a cell has been surveilled, its SAI value is reduced by calculating (28):

$$v_{cell\ x}(t) = \min[v_{\min}, \gamma_3 \times v_{cell\ x}(t-1)],\quad \gamma_3 < 1 \quad (28)$$

V. PROPOSED PSO-BASED TRAJECTORY PLANNER

In this section, we propose a trajectory planner for multiple UAVs based on standard PSO and the surveillance area priority updating mechanism explained in Section IV. The proposed planner first utilizes PSO to derive the optimal trajectories for each MDr, in which the proposed multi-objective fitness function is used to acquire the best flying waypoint sequence from all possible trajectories. Then CAC is exploited to detect the existence of collisions. If CAC is satisfied, the planner will generate the optimal trajectories for one flight time. Finally, the SAI values will be updated before the next flight. Details are explained in following subsections.

A. PARTICLE SWARM OPTIMIZATION

Particle swarm optimization is a widely used evolutionary heuristic search algorithm for solving optimization problems, and was initially proposed by Kennedy and Eberhart in 1995 [48]. In PSO, each particle corresponds to a candidate solution that is randomly initialized. Then, in each iteration, the velocity and position of each particle are renewed based on information about the previous velocity, the best position ever occupied by the particle (personal influence), and the best position ever occupied by any particle in the

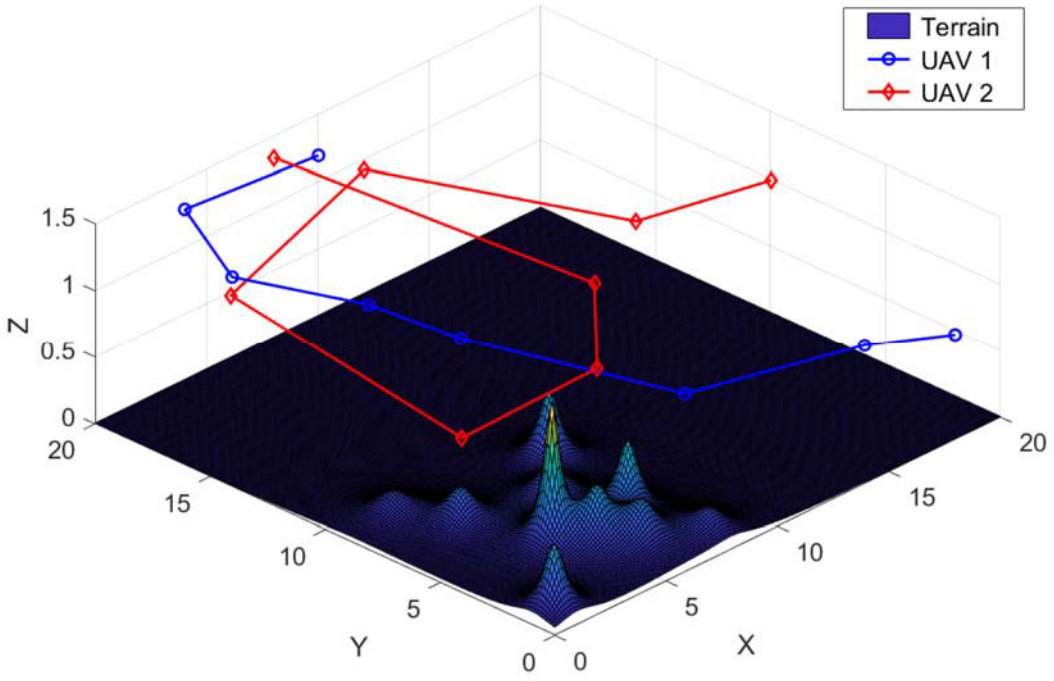


FIGURE 7. The optimal trajectories in a 3D representation.

swarm (social swarm). The mathematical formulations are as follows.

Assume the number of particles is P, the dimensionality of particles is D, and the iteration number is N. For the i-th particle, $x_i = (x_{i1}, x_{i2}, \ldots, x_{iD})$ and $v_i = (v_{i1}, v_{i2}, \ldots, v_{iD})$ represent the velocity and position vectors, respectively. For standard PSO, there are two kinds of cost value, i.e., individual best value, $P_{i,best}$, of one particle, and swarm best value, S_{best}, of all particles, which are depicted in (29):

$$\begin{aligned} P_{i,best} &= (p_{i1,best}, p_{i2,best}, \ldots, p_{iD,best}) \\ S_{best} &= (s_{1,best}, s_{2,best}, \ldots, s_{D,best}) \end{aligned} \tag{29}$$

Once the two cost values are determined, the velocity and position of each particle in each dimension are renovated by using (28).

$$\begin{aligned} v_{ij}^{k+1} &= \omega v_{ij}^{k} + c_1 r_1 (p_{ij,best} - x_{ij}^{k}) + c_2 r_2 (s_{j,best} - x_{ij}^{k}) \\ x_{ij}^{k+1} &= x_{ij}^{k} + v_{ij}^{k+1} \\ &\quad where \;\; i = 1, 2, \ldots, P \;\; j = 1, 2, \ldots, D \\ &\qquad\qquad k = 1, 2, \ldots, N \end{aligned} \tag{30}$$

In (30), r_1 and r_2 are random values between 0 and 1; ω is the inertia parameter, which reflects the influence of the velocity of the previous iteration on the current iteration; and c_1 andc_2 represent self-cognition and social knowledge, which indicate the inheriting abilities from the particle itself and from the whole swarm.

B. PROPOSED TRAJECTORY PLANNER

In our proposed planner, a feasible flight route consists of a sequence of waypoints and line segments. During our implementation, an eight-waypoint trajectory is adopted. We divide the entire operational area into unit cell areas.

The process of the proposed planner is shown in the algorithm pseudocode. At the beginning, we set estimated surveillance flight times to cover the whole operational area, and we initialize the SAI values for all cells. Next, we utilize PSO to derive the optimal trajectories for multiple MDrs, which corresponds to steps 5 to 33. During the optimization process, we first randomly generate the position and velocity vectors for N_{par} particles, and initialize $P_{t,best}$ and S_{best} as x_t and $x_{N_{par}}$, respectively. Next, the position and velocity vectors of each particle are renovated by using Formula (27). Then, the proposed multi-objective fitness functions are used to evaluate those newly updated particles, which consist of several objective and constraint functions explained in Section IV.

After that, we renew $P_{t,best}$ and S_{best} based on the fitness values of all the particles. Then, we store the optimal trajectory for the first MDr while the iteration number is set equal to N_{iter}. Finally, to avoid waypoint overlapping among multiple MDrs, i.e., repeated surveillance of the same area, we reduce the SAI values of those cells that have been surveilled before starting the next trajectory planning. Thus, the surveillance efficiency is improved. Since our research mainly focuses on collision-free path planning for multiple UAVs, the collision avoidance constraint is applied to determine if collisions could occur between MDr trajectories.

Algorithm 1 Pseudocode of the Proposed Multi-UAV Trajectory Planner

```
Set flight time = N_flight;
Initialize the SAI value of each cell;
for i = 1: N_flight
{
 while (CAC is not satisfied)
 {
  Set monitoring UAV number= N_MDr;
  for j = 1: N_MDr
  {
    Set iteration number= N_iter;
    Set particle number= N_par;
    for k = 1: N_iter
    {
     for t = 1: N_par
     {
      Randomly initialize x_t and v_t;
      Initialize P_{t,best} = x_t, S_best = x_{N_par};
      Update x_t and v_t using (30);
      Compute the fitness value of x_t using (3) to (22);
      if(fitness(x_t) > fitness(P_{t,best}))
      {P_{t,best} = x_t;}
      if (fitness(P_{t,best}) > fitness(S_best))
      {
       S_best = P_{t,best};
       Opt_fitness = fitness(S_best);
      }
     }
     }
    Update the SAI values using (28);
   }
   Evaluate collisions among multiple UAVs using (23);
 }
Generate the output of collision-free trajectories for multiple MDrs;
Update the SAI values using equations (24) to (27);
}
Generate the output of the N_flight flight times path planning results;
```

If CAC is not satisfied, we go back to Step 5. The optimal trajectories for multiple MDrs can be obtained only if CAC is satisfied. After that, SAI values should be renewed before executing the next flights by using formulas (22) to (25). Finally, when the flight time is equal to N_{flight}, it will generate all of the optimal planning trajectories for multiple MDrs to cover the whole operational area except for the restricted areas.

VI. SIMULATION RESULTS

In this section, we conducted a Matlab simulation for two MDrs over several flights times to evaluate the performance

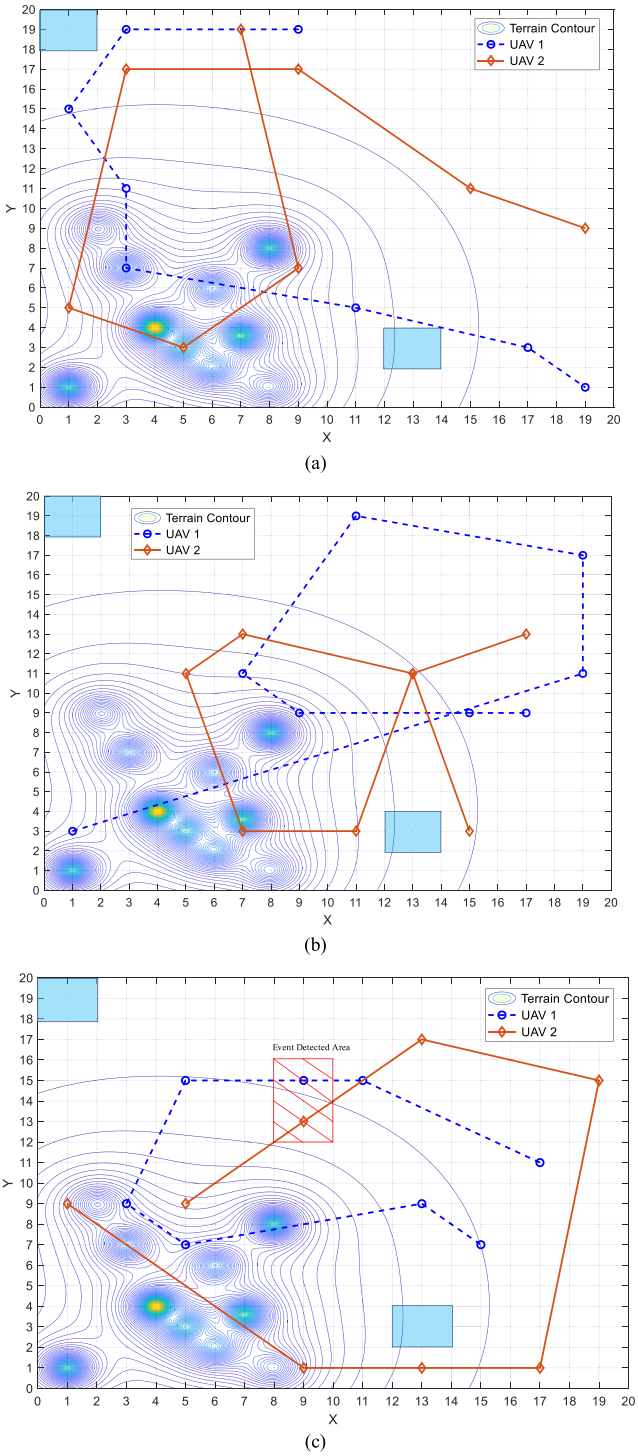


FIGURE 8. Trajectory planning results for (a) the first flight, (b) the second flight, and (c) the third flight.

of the proposed multiple-UAV trajectory planner. First, we evaluated the optimal trajectory solution generated by the proposed planner, and then, we verify the rationality of the proposed SAI updating mechanism based on the generated trajectories. For PSO parameter settings, Roberge *et al.* [19] suggested $c_1 = c_2 = 1.496$ and $\omega = 0.7298$. The main parameters used for the simulation are shown in Table 2.

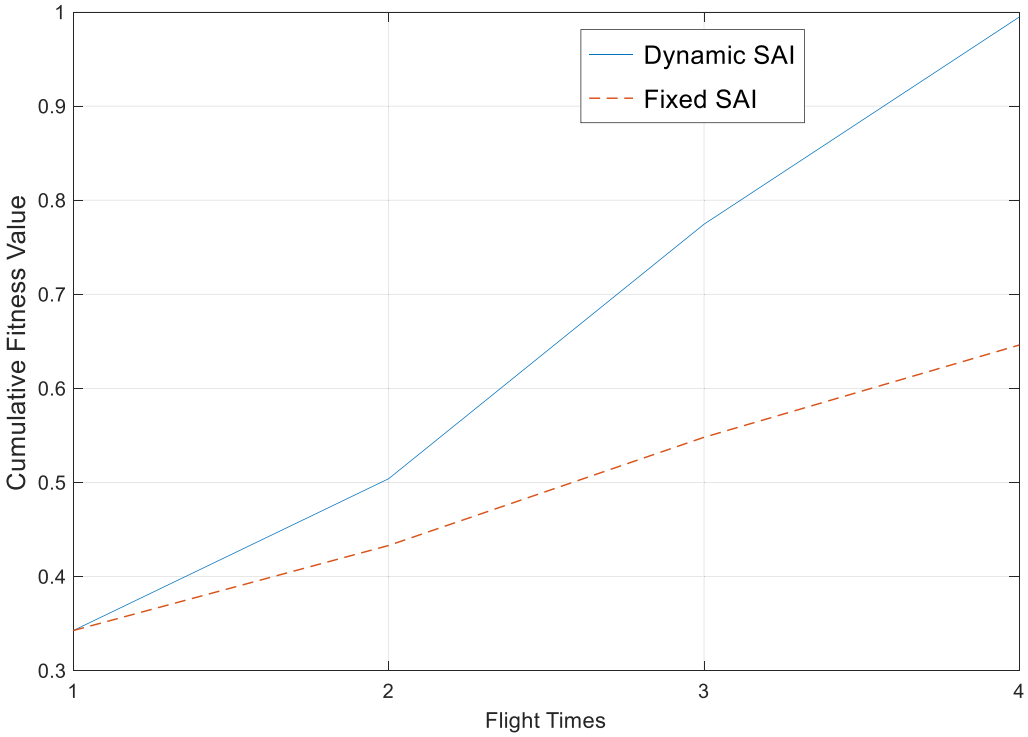


FIGURE 9. Cumulative fitness value comparison between the fixed SAI and the dynamic SAI.

TABLE 2. Main simulation parameters.

Parameters	Symbols	Values
Grid size	-	20×20
Operation space	-	20×20×1.5
Number of MDrs	N_{MD}	2
MDr unit power	P_u	20
MDr speed	v	10
Number of particles	N_{par}	32, 64, 128, 256, 512
Number of iterations	N_{iter}	50, 100
Event detection threshold	th_{event}	20
Flight time threshold	th_{flight}	2
SAI update control	γ_1, γ_2/γ_3	1.5/0.2
Maximum turning angle	θ_{max}	85°
Maximum gliding angle	α_{max}	-30°
Maximum climbing angle	β_{max}	30°
Minimal safe distance	d_{min}	0.2
Initial SAI value	v_{cellx}^{ini}	1 to 10
Initial environmental risk	r^e	1 to 5

In Figure 5, we show how the number of particles affects the optimal fitness value of the proposed fitness function. As expected, the fitness value converges to a stable value faster as the numbers of iterations and particles increase. Figure 6 shows the trajectory optimization performance of the first flight time in terms of energy consumption, flight risk, and surveillance area importance, in which the particle number is 256. As the number of iterations increases, the energy consumption and flight risk for the two MDrs minimize and maintain a stable value while their surveillance area importance values are maximized. Moreover, the SAI values of the optimal trajectories for the two MDrs are 191 and 187, while their energy consumption difference is less than five, which means the proposed trajectory planner can ensure fairness between generated trajectories. In our paper, we generate all the trajectories in a 3D environment as illustrated in Figure 7, which represents the optimal trajectory of the first flight.

In Figure 8, we show the surveillance trajectories of the two MDrs in accordance with event detection for the first three flight times. The forbidden areas and the event detection areas are marked by blue and red rectangles, respectively. Figure 8 (a) and (b) show the optimal trajectories for the two MDrs during the first and second flight time, respectively. We can see that the waypoints from the two flights do not overlap, which means that our proposed trajectory planner can surveil the operational area with high efficiency. We triggered two events at point (9,15) and (9,13) (i.e., the event-detection area) after the second flight. In Figure 8 (c), the MDrs visit those two points during the third flight time, which indicates that our proposed SAI updating mechanism can cover the operational area where the events happened.

For each flight time, we sum the optimal fitness values of the two MDrs. In Figure 9, we compare the cumulative

optimal fitness values between the fixed SAI and the dynamic SAI to validate the effectiveness of the proposed SAI updating method. As the flight time increases, the cumulative fitness value for dynamic SAI increases faster, indicating that our proposed updating SAI method can help MDrs visit the more important areas in a changing environment.

VII. CONCLUSION

In this paper, we propose a trajectory planner for multiple UAVs and apply it to MDrs to surveil a certain operational area to detect the existence of IDrs. To evaluate the trajectories generated by the proposed trajectory planner, we then introduce a multi-objective fitness function that has eight optimization indexes in terms of UAV maneuverability, energy consumption, flying risk, and surveillance area priority. The optimal trajectories are obtained by maximizing the fitness function values. Moreover, we also propose a surveillance area importance updating mechanism to effectively consider new events that happen in the operational area. The simulation results prove that our proposals can obtain collision-free trajectories for multiple UAVs with high fitness values, and they show highly dynamic environmental adaptability. Currently, we considered the two MDrs over several flights times to evaluate the performance of the proposed multiple-UAV trajectory planner. We meant to figure out the feasibility of the three dimension optimal surveillance trajectory planning for multiple UAVs by using PSO with surveillance area priority. We will consider more than two MDrs in our future work because our proposal can effectively perform to achieve the collision-free trajectories for multiple UAVs with high fitness values and adapting the highly dynamic environment.

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